

Hide menu

- Approximation Algorithms and Approximation Ratios: Introduction
- Jobshop Minimum Makespan Scheduling
- Vertex Cover
- Video: Approximation Algorithms for Vertex Cover and their Analysis31 min
- Lab: Interactive Notes on Vertex Cover Approximation Algorithms1h
- Quiz: Vertex CoverSubmitted
- Maximum Satisfiability Problem
- Problem Set # 3

Vertex Cover

Review Learning Objectives

Submit your assignment

Due

Feb 25, 11:59 PM IST

Receive grade

To Pass

80% or higher

Like

Dislike

Report an issue

Congratulations! You passed!

Grade

received 100%

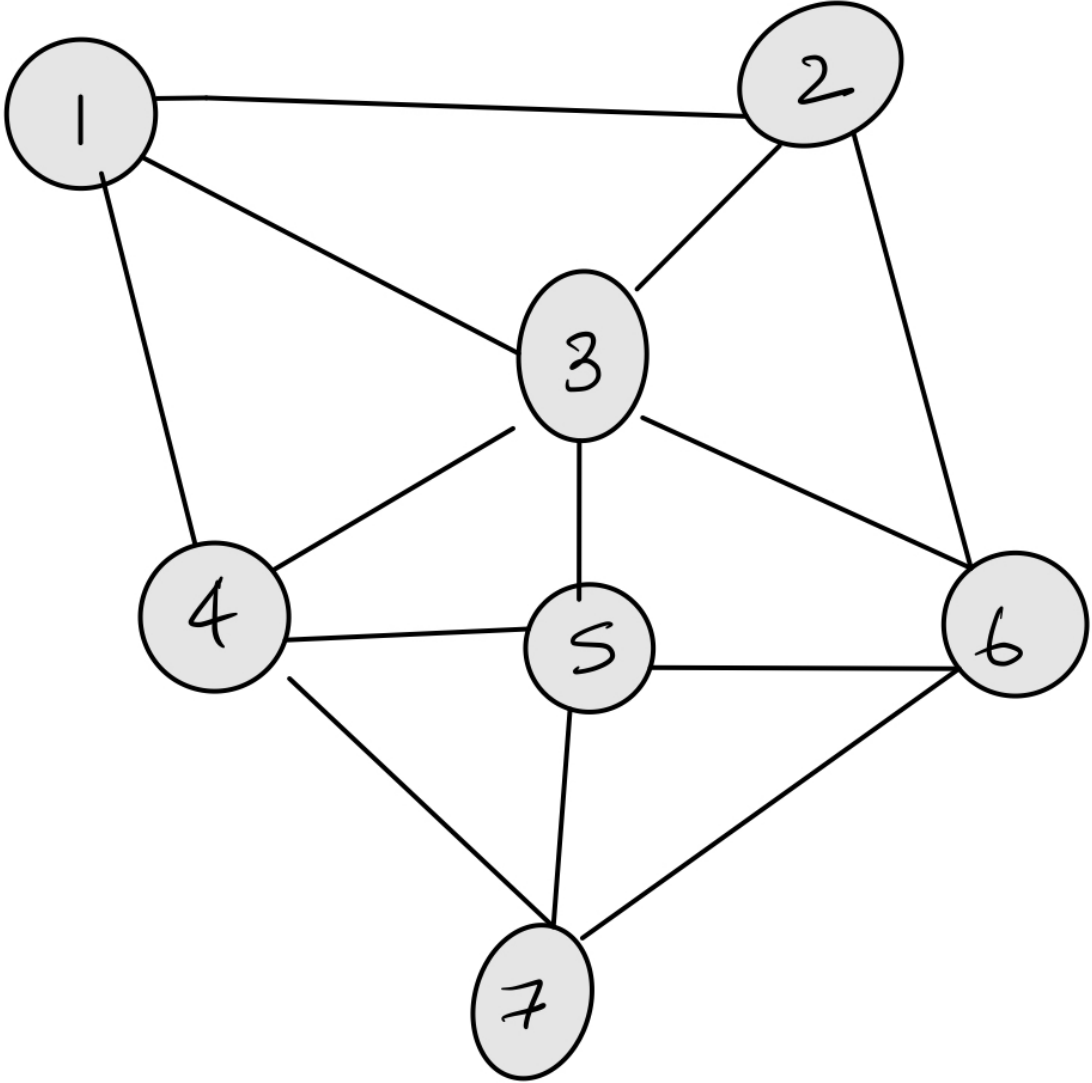
Latest Submission

Grade 100%

To pass 80% or higher

Go to next item

The questions below concern the following graph with 7 nodes.



Try again

Your grade

100%

View Feedback

We keep your highest score

1. Suppose we apply the greedy algorithm that repeatedly chooses the maximum degree node in the graph, adds it to the cover and then deletes it from the graph. Note that we break ties between multiple vertices that have the same degree by choosing the vertex with lowest number/ID.

4 / 4 points

☒ The greedy algorithm produces a vertex cover of size 5 that involves nodes 1, 3, 4, 5, 6

Correct

Correct

☐ The greedy algorithm produces the cover 1, 4, 5, 6 of size 4 and is in fact optimal.

☒ Node 3 is the very first node added to the cover.

Correct

Correct: its degree is 5, which is more than any other node in the graph.

☒ Node 4 is the second node added to the cover.

Correct

After node 3 is deleted as the first node, nodes 4, 5, 6, 7 each will have degree 3 but we choose node 4 since by convention we choose lowest index node to break ties.

☐ Node 1 is the second node added to the cover.

☒ Node 6 is the third node added to the cover.

Correct

Correct: the first node would be node 3, after which we add node 4 and then node 6 since it has the highest degree after nodes 3, 4 are deleted.

2. Suppose we ran the greedy algorithm based on maximal matching and processed the edges in the following order:

4 / 4 points

$[(1, 2), (1, 3), (1, 4), (2, 3), (2, 6), (3, 4), (3, 5), (3, 6), (4, 5), (4, 7), (5, 6), (5, 7), (6, 7)]$

What is the size of the vertex cover that we obtain?

6

Correct

Check that the algorithm adds the vertices 1, 2, 3, 4, 5 and 6

